PALINDROMIC-PREFIX XOR FEEDBACK: ONE TWO-CYCLE, A SHARP LINEAR CLOCK, AND EXACT FIBRES

ANONYMOUS

Abstract. For a binary word, form the complete indicator vector of its palindromic prefixes and XOR that vector with the word. We iterate this synchronous self-map. Global complement removes a phase bit and produces a normalized system in which one update zeros the first three coordinates and every later update extends a leading zero prefix. We prove that the original system has the single strict two-cycle $0^n \leftrightarrow 1^n$ and that its maximum transient depth is 0 at length one, 1 at length two, and sharply $n-2$ thereafter. A periodic source whose ones occur at positions congruent to 3 modulo 4 attains the bound. We also give a complete left-to-right decoder for every one-step fibre; it is simultaneously an exact image test. Palindrome recognition and static palindrome data structures are credited background. Computation is used only for exact finite checks. External release remains on hold.

1. The feedback map and the subtraction boundary

For a word $x = x_1 \cdots x_n \in \{0,1\}^n$, put

$$p_i(x) = \mathbf{1}\{x_1 \cdots x_i = x_i \cdots x_1\}, \qquad 1 \le i \le n.$$

All arithmetic on bits is in $\mathbb{F}_2$, written $\oplus$. We study the self-map

$$T_n(x)_i = x_i \oplus p_i(x). \tag{1}$$

The entire prefix-indicator vector is recomputed after every update.

Initial-palindrome recognition has real-time algorithms [2]; palindromic trees support access to all palindromic factors [4], and recent work compresses families of palindromic prefixes [1]. Static palindromic generation of binary words has a separate structural theory [3]. These results, ordinary border algorithms, and generic XOR-network terminology are background here. Our object is only the repeated full-vector feedback in (1), its temporal dynamics, and its inverse fibres.

A state is *recurrent* if it belongs to a directed cycle. Its depth is the least number of updates needed to reach a recurrent state. Write $\bar{x} = x \oplus 1^n$ for global complement and normalize a word by

$$N(x)_i = x_i \oplus x_1. \tag{2}$$

Thus $N(x)_1 = 0$, and the two words in a complement pair have the same normalization.

2. Complement quotient and the recurrent class

Proposition 2.1 (exact quotient). *For every $x \in \{0,1\}^n$,*

$$T_n(\bar{x}) = \overline{T_n(x)}.$$

On the normalized carrier $\mathcal{Q}_n = \{y \in \{0,1\}^n : y_1 = 0\}$, the induced map is

$$Q_n(y)_i = y_i \oplus 1 \oplus p_i(y), \tag{3}$$

and $N(T_n(x)) = Q_n(N(x))$.

Proof. Complementing every letter preserves every equality between two positions, so $p_i(\bar{x}) = p_i(x)$. Equation (1) then gives the first identity. Normalization also preserves all equality tests, hence $p_i(N(x)) = p_i(x)$. Since $p_1(x) = 1$,

$$N(T_n(x))_i = (x_i \oplus p_i(x)) \oplus (x_1 \oplus 1) = N(x)_i \oplus 1 \oplus p_i(N(x)),$$

which is (3). At $i = 1$ its value is zero, so Q_n is closed on $\mathcal{Q}_n$. □

For $y \in \mathcal{Q}_n$, let $\ell(y)$ be the length of its leading zero run.

Lemma 2.2 (reset and amplifier). *For every $y \in \mathcal{Q}_n$, the first $\min(3,n)$ coordinates of $Q_n(y)$ are zero. More generally, if y begins with 0^k and $1 \le k < n$, then $Q_n(y)$ begins with 0^{k+1}.*

2020 *Mathematics Subject Classification.* 37P25, 68R15, 05A15.
Key words and phrases. finite dynamical system, palindrome, binary word, synchronization, fibre.

Proof. The first coordinate is zero by (3). Write the first three source bits as $0ab$ when they exist. The length-two prefix is palindromic exactly when $a = 0$, and the length-three prefix is palindromic exactly when $b = 0$. Thus at each of coordinates two and three, the last two terms in (3) cancel the source bit.

Now suppose that $y_1 \cdots y_k = 0^k$. Every prefix ending at or before k is palindromic, so its image bit is zero. If $y_{k+1} = 0$, the next prefix is also all zero. If $y_{k+1} = 1$, its first and last bits differ, so it is not palindromic. In either case (3) gives $Q_n(y)_{k+1} = 0$. □

Theorem 2.3 (complete recurrence and upper clock)**.** *The only recurrent class of T_n is the strict two-cycle*

$$0^n \longleftrightarrow 1^n.$$

The maximum depth is at most 0 for $n = 1$, at most 1 for $n = 2$, and at most $n - 2$ for $n \geq 3$.

Proof. The normalized word 0^n is fixed by Q_n. If a normalized word is not zero, then after one update it has at least $\min(3, n)$ leading zeros by lemma 2.2; every subsequent nonzero state has a strictly longer leading zero prefix. Hence 0^n is the only recurrent quotient state. The number of required updates is zero at $n = 1$, at most one at $n = 2$, and at most $1 + (n - 3) = n - 2$ for $n \geq 3$.

In the original system the first bit flips at every update because $p_1(x) = 1$. Once the normalization is zero, the word is constant, and every prefix of a constant word is palindromic. Therefore $T_n(0^n) = 1^n$ and $T_n(1^n) = 0^n$. Each recurrent original state must project to the unique recurrent quotient state, and the phase flip then forces exactly this strict two-cycle. □

3. A sharp periodic source

For $n \geq 3$, define $u^{(n)} \in \mathcal{Q}_n$ by

$$u_i^{(n)} = 1 \quad \Longleftrightarrow \quad i \equiv 3 \pmod 4. \tag{4}$$

For $3 \leq k \leq n$, let $v_k^{(n)}$ be the length-n prefix of $0^k 1010 \cdots$; in particular $v_n^{(n)} = 0^n$.

Lemma 3.1 (exact sharp trajectory)**.** *For $n \geq 3$,*

$$Q_n(u^{(n)}) = v_3^{(n)}, \qquad Q_n(v_k^{(n)}) = v_{k+1}^{(n)} \quad (3 \leq k < n).$$

Consequently $u^{(n)}$ reaches 0^n for the first time after $n - 2$ updates.

Proof. The palindromic prefixes of $u^{(n)}$ have lengths 1, 2, and the lengths $i \geq 5$ satisfying $i \equiv 1 \pmod 4$. To verify this, for $i \geq 3$ note that the one-positions in the prefix are $3, 7, \ldots$. Reflection in a word of length i sends position $3 + 4j$ to $i - 2 - 4j$. This preserves that set exactly when $i \equiv 1 \pmod 4$; in every other residue class the reflection of position 3 is not a one-position. Substitution in (3), one residue class at a time, gives zeros at positions one through three and thereafter a one exactly at each even position. This is $v_3^{(n)}$.

Fix $k \geq 3$. The prefixes of $v_k^{(n)}$ of length at most k are all zero and hence palindromic. No longer prefix is palindromic. Indeed, such a palindrome would have to end in 0^k. If its length is less than $2k$, its last k positions include position $k + 1$, whose bit is one. If its length is at least $2k$, its last k positions lie in an alternating tail and again contain a one. Thus beyond position k the palindrome indicator vanishes, and (3) complements the alternating tail. Its first one is thereby absorbed into the zero prefix, giving exactly $v_{k+1}^{(n)}$.

The displayed chain contains $n - 2$ arrows from $u^{(n)}$ to $v_n^{(n)}$. Every preceding member is nonzero, so this is the first hitting time. □

Corollary 3.2 (sharp clock)**.** *The maximum depth of T_n is*

n	1	2	$n \geq 3$
max depth	0	1	$n - 2$.

Proof. The upper bounds are theorem 2.3. At length two either nonconstant word has nonzero normalization and reaches the recurrent class in one step. For $n \geq 3$, choose either original phase above the normalized word $u^{(n)}$; lemma 3.1 gives depth $n - 2$. □

4. The complete one-step fibre decoder

We now count the fibre of an arbitrary original target $t \in \{0,1\}^n$. Put $z = N(t)$. Define sets $\mathcal{D}_i(z)$ of normalized source prefixes recursively. Start with $\mathcal{D}_1(z) = \{0\}$. Given $y_1 \cdots y_{i-1} \in \mathcal{D}_{i-1}(z)$, let $m = y_2 \cdots y_{i-1}$, where the empty word is palindromic. Extend by the following rule:

(5)

m	z_i	allowed y_i
nonpalindromic	0 or 1	$1 \oplus z_i$
palindromic	0	$0, 1$
palindromic	1	none.

Theorem 4.1 (every-target fibres)**.** *For every $t \in \{0,1\}^n$,*

$$|T_n^{-1}(t)| = |\mathcal{D}_n(N(t))|.$$

In particular, t belongs to the image exactly when the decoder leaves at least one branch. The original complement phase contributes no additional multiplicity.

Proof. Consider a normalized source prefix $y_1 \cdots y_i$, with $y_1 = 0$. Its full length-i prefix is palindromic exactly when $y_i = 0$ and the middle word $m = y_2 \cdots y_{i-1}$ is palindromic. If m is nonpalindromic, the indicator in (3) is zero for either y_i, so the target equation $Q_n(y)_i = z_i$ forces $y_i = 1 \oplus z_i$. If m is palindromic, then $y_i = 0$ turns the indicator on and $y_i = 1$ turns it off; both choices give target bit zero, and neither gives target bit one. These are precisely the three cases in (5).

The ith quotient output depends only on the first i source bits. Induction on i therefore shows that $\mathcal{D}_i(z)$ is exactly the set of normalized length-i prefixes whose quotient image agrees with z through position i. At $i = n$ it is the complete normalized fibre.

Finally, if x is an original source of t, then its first bit is forced by $t_1 = x_1 \oplus 1$, namely $x_1 = 1 \oplus t_1$. Every normalized branch y therefore lifts to the unique source $x_i = y_i \oplus (1 \oplus t_1)$, and proposition 2.1 shows that this source maps to t. Hence there is no factor of two. □

Remark 4.2. The decoder is target-wise, including targets outside the image. It is not merely a recurrence for the largest fibre: it returns the exact cardinality of every fibre using only palindrome tests on prefixes already constructed.

5. Exact control and limitations

The dependency-free paper-local verifier constructs every functional graph for $1 \le n \le 18$, totaling 524,286 states. It checks complement equivariance, the literal quotient, all cycles and depths, and both amplifier statements. For every target through $n = 15$, it independently compares (5) with the literal fibre. It also checks the closed sharp family through $n = 64$. The canonical run executes 3,870,590 exact assertions.

These finite checks are counterexample pressure and do not prove statements at untested lengths. Conversely, no observed image or maximum-fibre sequence is promoted to an all-length claim. The owner search is bounded, and a non-hit is not a novelty or priority certificate. This anonymous artifact is therefore marked `HOLD_EXTERNAL`: posting, submission, and release are outside the present result.

References

[1] Gabriel Bathie, Jonas Ellert, and Tatiana Starikovskaya. Small space encoding and recognition of k-palindromic prefixes. In *36th International Symposium on Algorithms and Computation (ISAAC 2025)*, volume 359 of *Leibniz International Proceedings in Informatics (LIPIcs)*, pages 9:1–9:16. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2025. doi: 10.4230/LIPIcs.ISAAC.2025.9.

[2] Zvi Galil. Palindrome recognition in real time by a multitape Turing machine. *Journal of Computer and System Sciences*, 16(2):140–157, 1978. doi: 10.1016/0022-0000(78)90042-9.

[3] Tero Harju, Mari Huova, and Luca Q. Zamboni. On generating binary words palindromically. *Journal of Combinatorial Theory, Series A*, 129:142–159, 2015. doi: 10.1016/j.jcta.2014.10.003.

[4] Mikhail Rubinchik and Arseny M. Shur. EERTREE: An efficient data structure for processing palindromes in strings. *European Journal of Combinatorics*, 68:249–265, 2018. doi: 10.1016/j.ejc.2017.07.021.